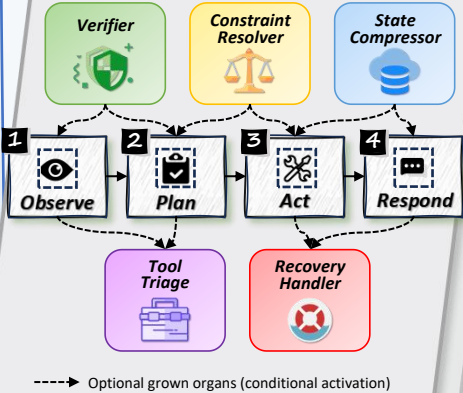


HarnessGen: Diagnosis-Triggered Structural Adaptation of LLM Agent Harnesses

1. Execution Loop

Task batch sampled from $p(\tau)$

Current harness (body plan) as DAG



Active subgraph per step (topological execution)

Trajectories

Failure Set $\mathcal{F}_k$

Structured descriptors $\psi(\xi)$:
topology, timing, state, tool, outcome

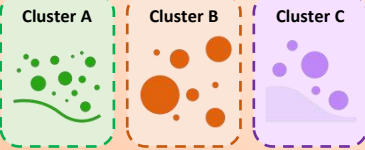
2. Failure Diagnosis

Failure trajectories $\mathcal{F}_k$
(from execution loop)

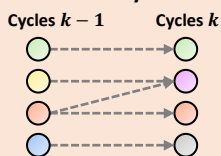
Reduced descriptors $\psi(\xi)$

PCA to 32D

DPM clustering



Hungarian matching across cycles



persistent for $\geq k$ cycles

Each persistent cluster c enters recalibration test on existing organs only.

Prompt Revision

Trigger threshold adjustment

Routing weight update

Few-shot selection

Recalibration data split

50% calibration

25% validation

25% diagnostic holdout

After bounded recalibration, performance $\geq \epsilon_{struct}$

YES

Miscalibration failure
→ keep topology
update parameters θ only.

NO

Residual gap $\Delta_c > 0$ and
persistent cluster

Missing-organ candidate

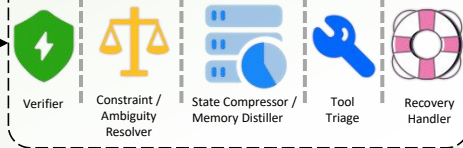
3. Bounded Recalibration Gate

next development cycle

Developmental controller

conditional on failure summary + current topology + residual gap

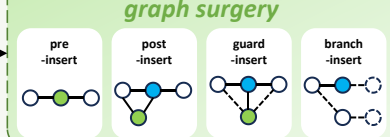
Typed schemalibrary (families)



Organ proposal ρ_c

- candidate organ
- activation rule ϕ
- insertion vertex v_c
- task scope $\mathcal{S}_c$
- lifecycle policy Λ_c

Schema-constrained graph surgery



Sandboxed replay validation



Utility gain
Latency cost
= $Utility(\rho_c)$

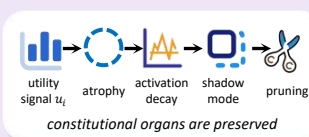
Accept if utility $> \eta$
and regression $\leq \epsilon_{reg}$

YES

Updated body plan $\mathcal{B}_{k+1}$

NO

Lifecycle management (grown organs only)



Structural budget (limit bloat)

4. Structural Adaptation



New organs are proposed only after persistent failures survive bounded recalibration.
HarnessGen distinguishes miscalibration (fix parameters) from missing capability (grow structure).

→ solid = data flow
... → dashed = control/selection (lifecycle signals)